Sheet 1

File: memory-v2.0-sheet1.kicad_sch

Sheet 2

File: memory-v2.0-sheet2.kicad_sch

Sheet 3

File: memory-v2.0-sheet3.kicad_sch

Sheet 4

File: memory-v2.0-sheet4.kicad_sch

Sheet 5

File: memory-v2.0-sheet5.kicad_sch

Sheet 6

File: memory-v2.0-sheet6.kicad_sch

Sheet 7

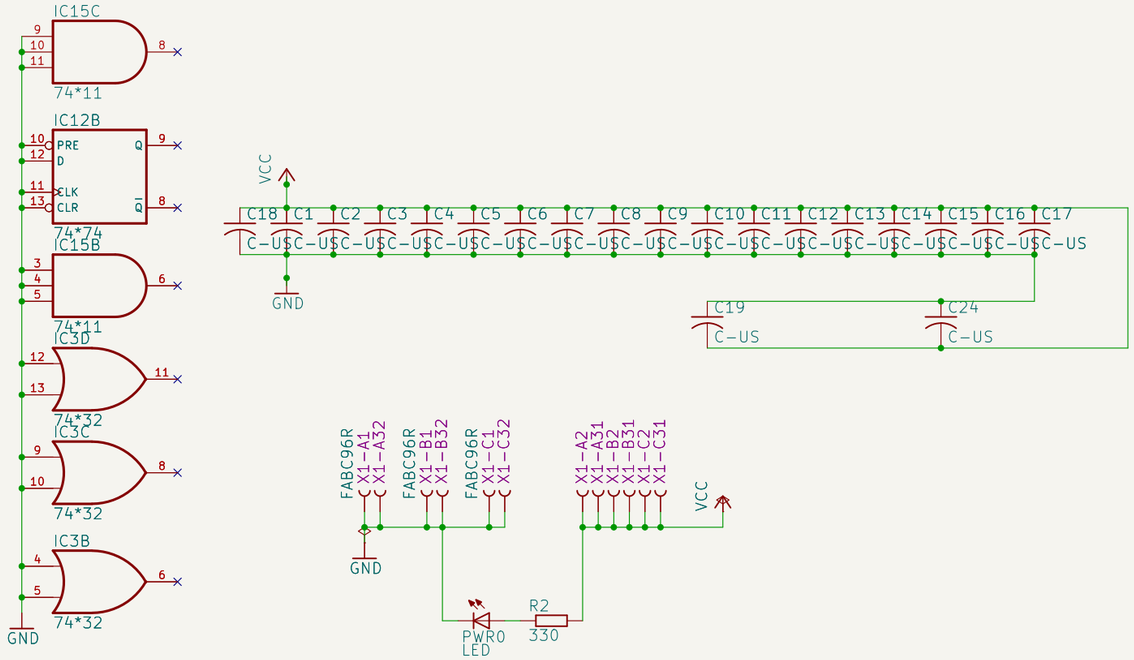
File: memory-v2.0-sheet7.kicad_sch

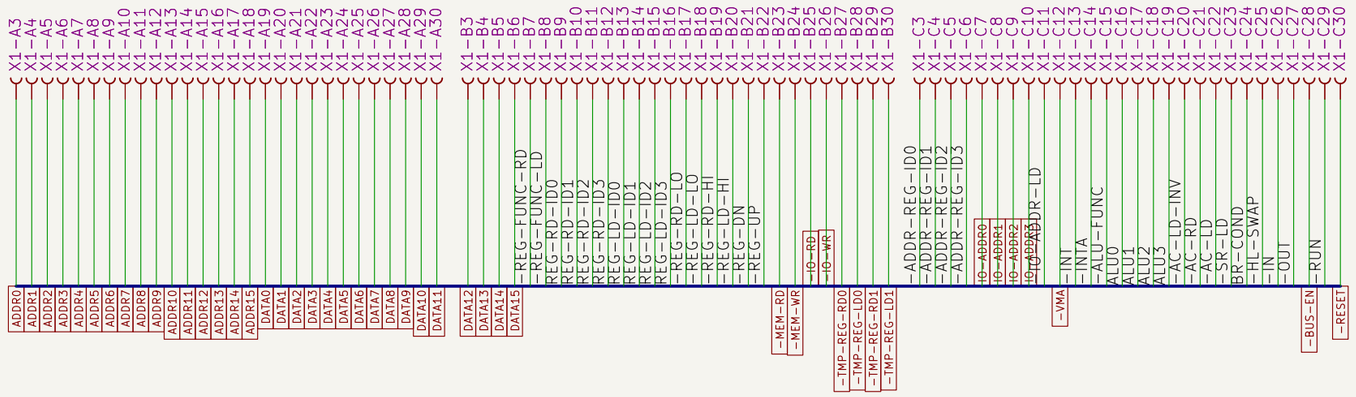
Circuit source: mem_v2_netlist.py; generated by gen_mem_v2.py; see README.md
Sheets 1-6 = memory card v1.3 as built (sheet 1: six bus labels made global); sheet 7 = CompactF

Sheet: /
File: memory-v2.0.kicad_sch

Title: YACC1 memory card v2.0

Size: A4	Date: 2026-09-24	**Rev: v2.0**
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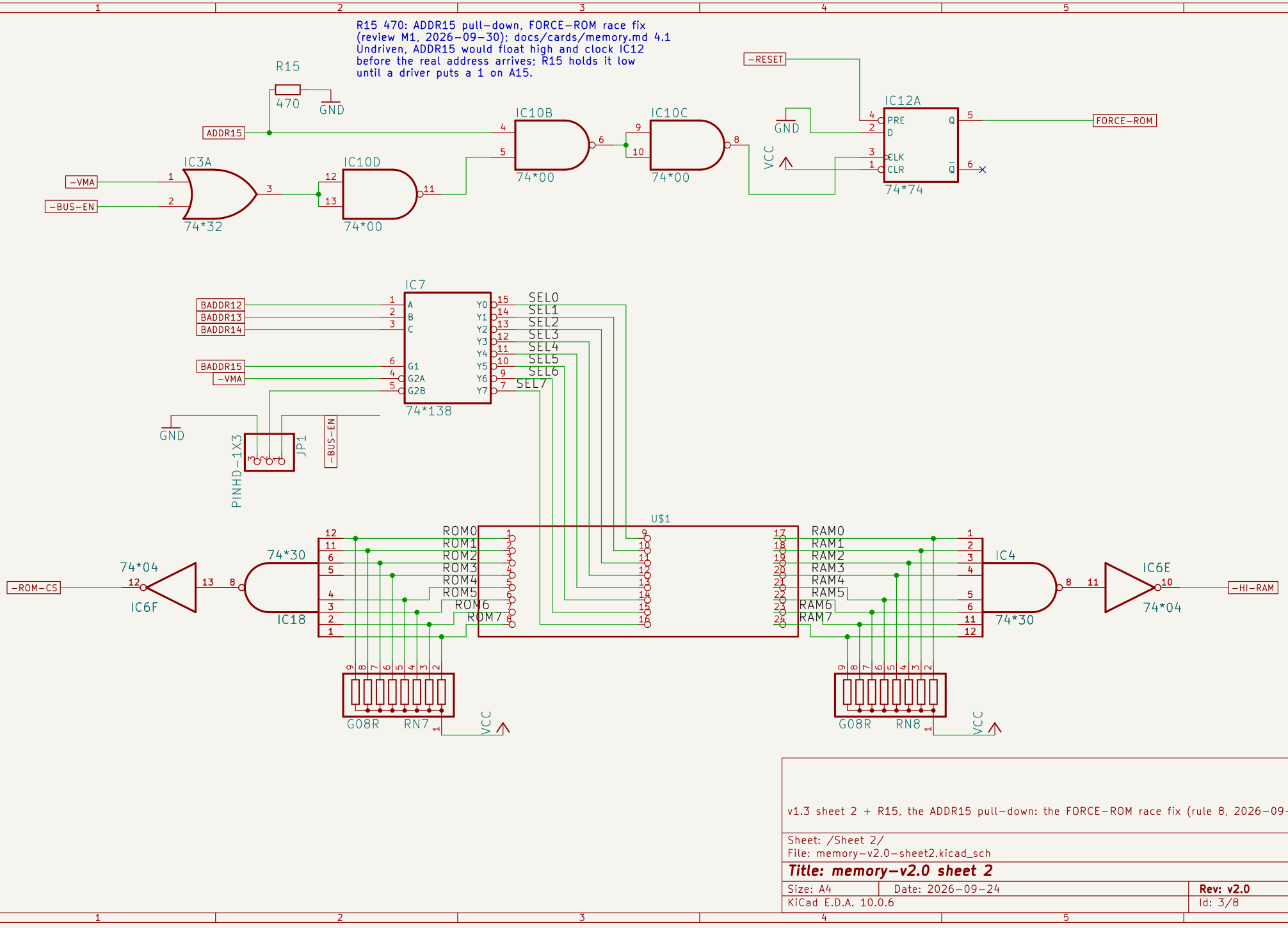
Memory card v2.0: IO-ADDR0-3 (C7-C10), -IO-RD (B25), -IO-WR (B26) and -RESET (C30) also feed the CompactFlash section on sheet 7 (I/O ports P8 = register latch, P9 = data; its own 74LS138 IC30, enabled by IO-ADDR3). DATA0-7 reach its data buffer IC34 and latch IC32. v1.3 used none of the I/O signals: the six labels above were X1-only local labels and are global labels now.
C20-C23 (v1.3: four 100 nF with no IC beside them, in the cap row between C19 and C24) are removed on v2.0 (2026-09-24); every IC keeps its own 100 nF.

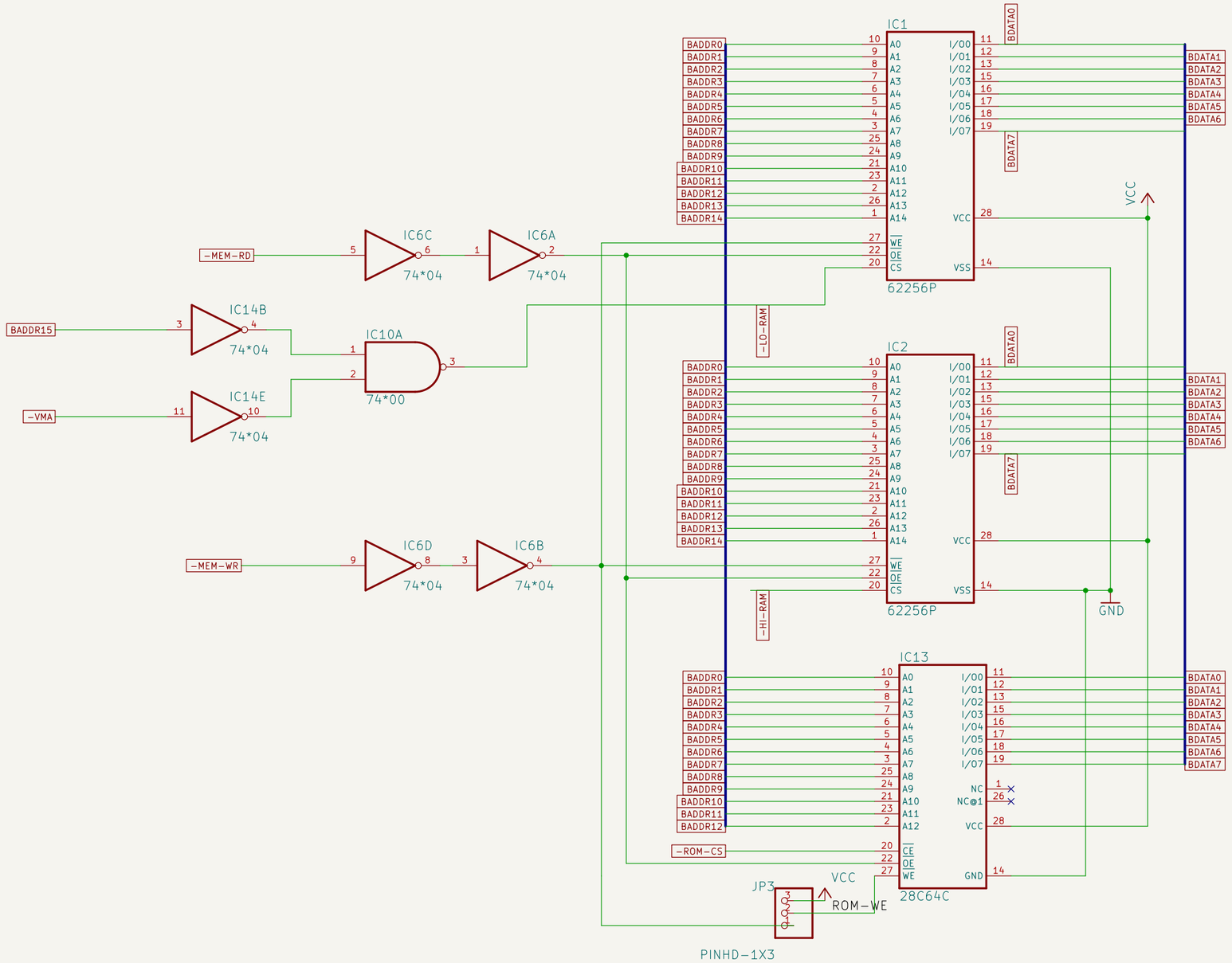
v1.3 sheet 1: 6 bus labels global (CF, sheet 7), C20-C23 removed, + note

Sheet: /Sheet 1/
File: memory-v2.0-sheet1.kicad_sch

Title: memory-v2.0 sheet 1

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R15 470: ADDR15 pull-down, FORCE-ROM race fix
(review M1, 2026-09-30); docs/cards/memory.md 4.1
Undriven, ADDR15 would float high and clock IC12
before the real address arrives; R15 holds it low
until a driver puts a 1 on A15.
R15
470
GND
ADDR15
IC10B
74*00
IC10C
74*00
-RESET
IC12A
PRE
D
CLK
CLR
Q
Q
74*74
FORCE-ROM
VCC
IC3A
74*32
-VMA
-BUS-EN
IC10D
74*00
IC7
74*138
BADDR12
BADDR13
BADDR14
BADDR15
-VMA
A
B
C
G1
G2A
G2B
Y0
Y1
Y2
Y3
Y4
Y5
Y6
Y7
SEL0
SEL1
SEL2
SEL3
SEL4
SEL5
SEL6
SEL7
GND
PINHD-1X3
JP1
-BUS-EN
U$1
ROM0
ROM1
ROM2
ROM3
ROM4
ROM5
ROM6
ROM7
RAM0
RAM1
RAM2
RAM3
RAM4
RAM5
RAM6
RAM7
-ROM-CS
74*04
IC6F
74*30
IC18
IC4
74*30
IC6E
74*04
-HI-RAM
G08R
RN7
VCC
G08R
RN8
VCC
v1.3 sheet 2 + R15, the ADDR15 pull-down: the FORCE-ROM race fix (rule 8, 2026-09-30)
Sheet: /Sheet 2/
File: memory-v2.0-sheet2.kicad_sch
Title: memory-v2.0 sheet 2
Size: A4
Date: 2026-09-24
Rev: v2.0
KiCad E.D.A. 10.0.6
Id: 3/8



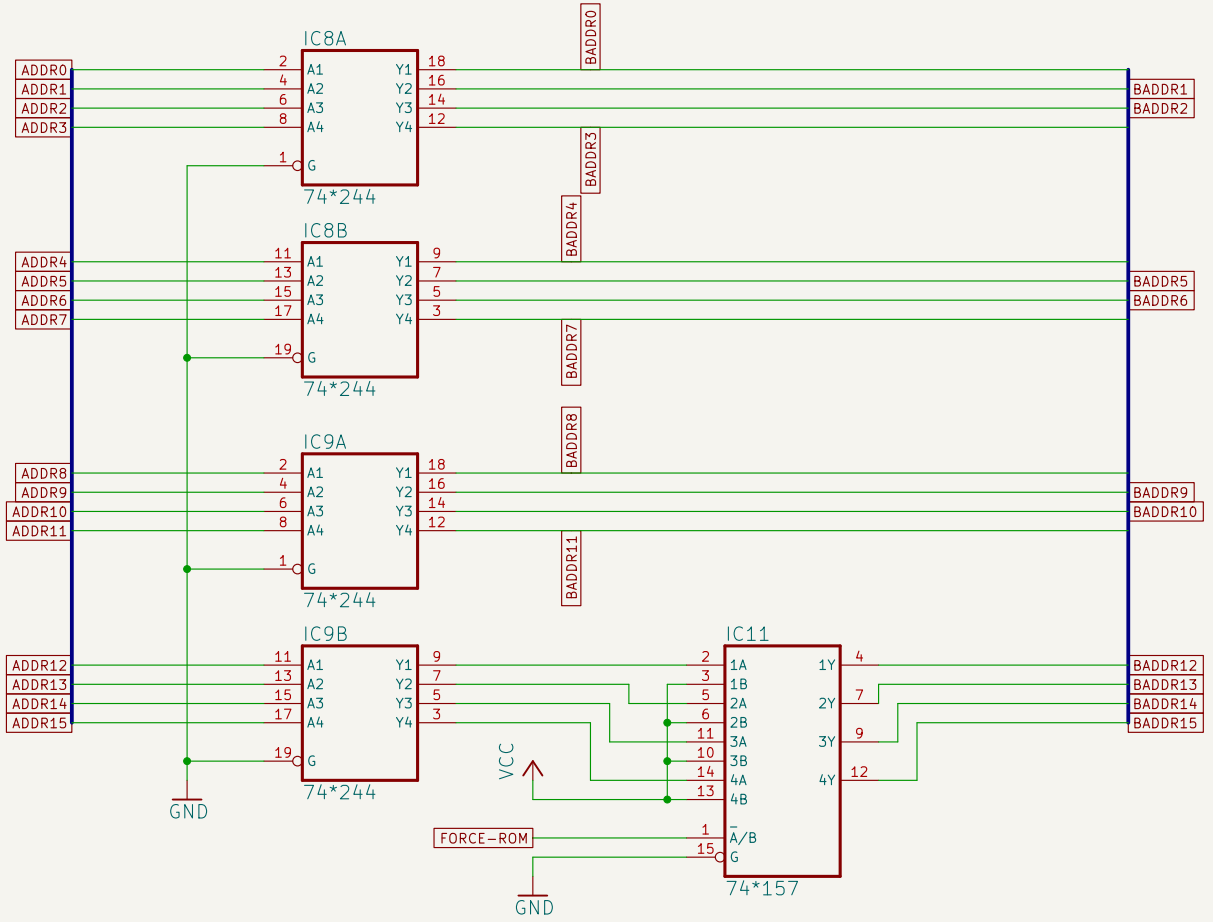
JP3 (v2.0, 2026-09-29): ROM write protect (design review M2).
1-2 = WRITE: IC13 -WE = -MEM-WR, as on v1.3. 2-3 = PROTECT: -WE held at VCC.
Fitted on 2-3: burn the 28C64 in a programmer; 1-2 only to write it in place.

v1.3 sheet 3 + JP3, the ROM write-protect jumper on IC13 -WE (rule 7, 2026-09-29)

Sheet: /Sheet 3/
File: memory-v2.0-sheet3.kicad_sch

Title: memory-v2.0 sheet 3

Size: A3	Date: 2026-09-24	**Rev: v2.0**
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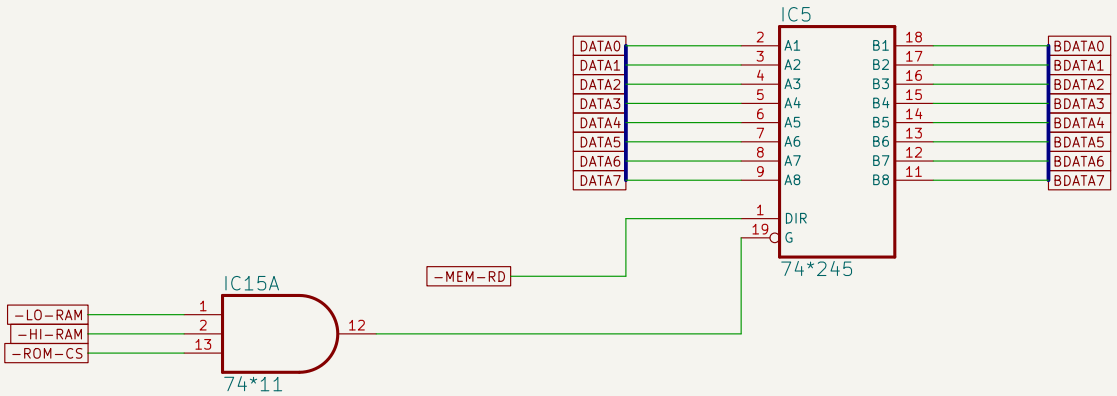


Unchanged from the built v1.3 (Fusion export 2026-09-24, Memory V1.3.sch sheet 4)

Sheet: /Sheet 4/
File: memory-v2.0-sheet4.kicad_sch

Title: memory-v2.0 sheet 4

Size: A4	Date: 2026-09-24	**Rev: v2.0**
KiCad E.D.A. 10.0.6		Id: 5/8

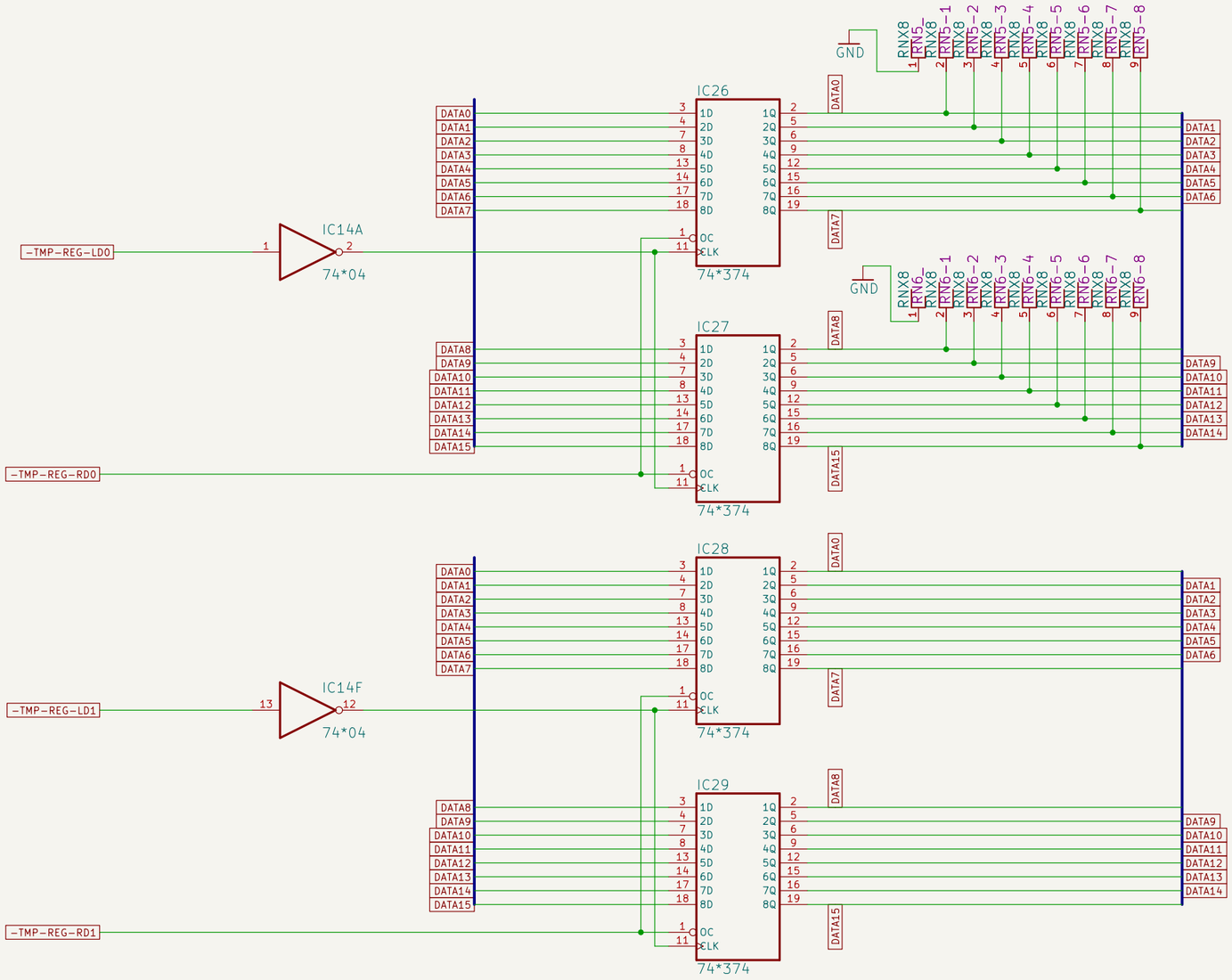


Unchanged from the built v1.3 (Fusion export 2026-09-24, Memory V1.3.sch sheet 5)

Sheet: /Sheet 5/
File: memory-v2.0-sheet5.kicad_sch

Title: memory-v2.0 sheet 5

Size: A4	Date: 2026-09-24	**Rev: v2.0**
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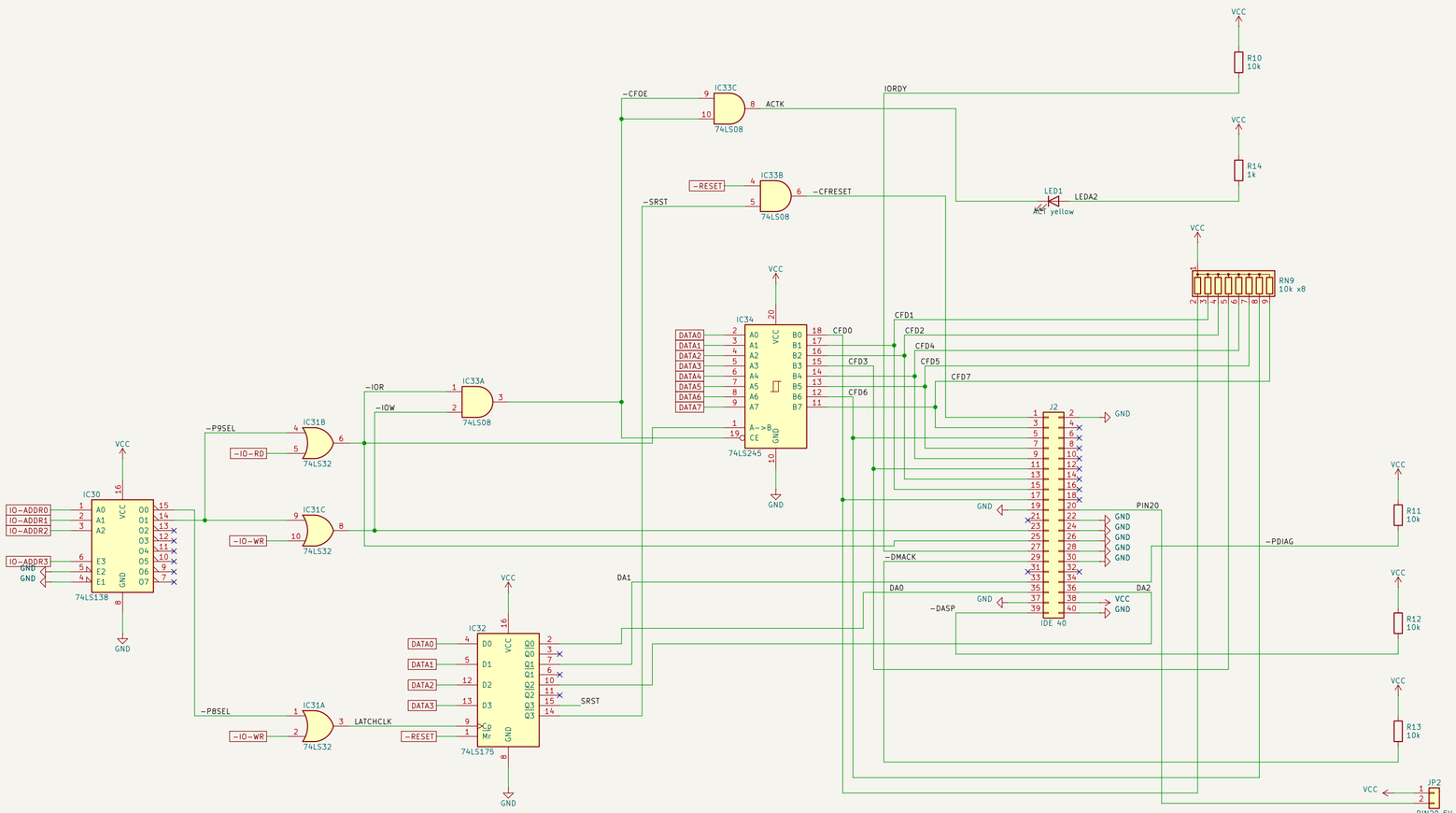
Unchanged from the built v1.3 (Fusion export 2026-09-24, Memory V1.3.sch sheet 6)

Sheet: /Sheet 6/
File: memory-v2.0-sheet6.kicad_sch

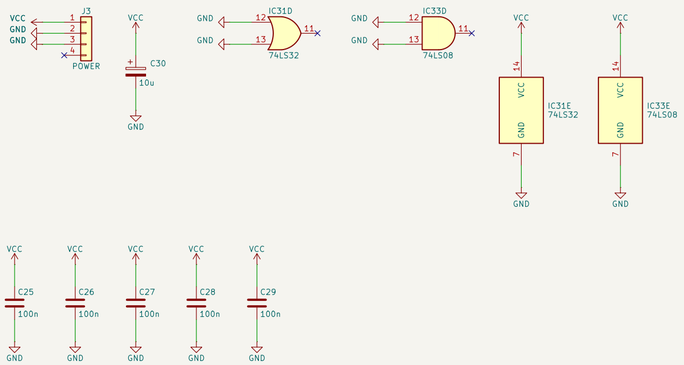
Title: memory-v2.0 sheet 6

Size: A3	Date: 2026-09-24	**Rev: v2.0**
KiCad E.D.A. 10.0.6		Id: 7/8

MEMORY CARD v2.0: THE COMPACTFLASH SECTION (I/O PORTS P8 / P9)



ADAPTER POWER J3, UNUSED GATES, POWER UNITS, DECOUPLING



DESIGN NOTES (MEMORY CARD v2.0, CF SECTION)

The CF section has no bus connector of its own: it shares the memory card's X1. Boxed global labels = nets shared with sheets 1–6: IO-ADDR0–3 (X1 C7–C10) –> IC30; -IO-RD (B25) –> IC31 gate 2 (CF -IOR); -IO-WR (B26) –> IC31 gates 1 and 3 (latch clock, CF -IOW); -RESET (C30) –> IC32 CLR, IC33 gate 2 (also IC12 PRE); DATA0–7 (A19–A26) –> IC34 A side. DATA0–3 –> IC32 D1–D4. On v1.3, IO-ADDR0–3, -IO-RD and -IO-WR reached X1 only. The CF section drives the data bus only while the CPU reads port P9 (IC34 enabled by -CFOE, DIR = -IOR).

1. Two I/O ports. P8 (write): 74LS175 latch IC32 – bits 0–2 = the ATA task-file register (CF DA0–2), bit 3 = CF reset (1 = held in reset). P9 (read/write): the selected ATA register, 8-bit True IDE, through the 74LS245 IC34.
2. Decode: 74LS138 IC30 on IO-ADDR0–2, enabled by IO-ADDR3 (Y0 = port 8, Y1 = port 9), as on the CF card v1.0. The ROM in the machine and both emulators use P8/P9. The I/O card must keep its IO-ADDR-HL strap at P0–P7.
3. IC31 (74LS32) ORs each port select with -IO-WR / -IO-RD: -IOR, -IOW and the latch clock are low only during the bus strobe for this port. -CS0 (J2 pin 37) LOW and -CS1 (38) HIGH: the strobes alone define every cycle (docs/cards/cf.md section 4); gating -CS0 would break the chip-select hold time after -IOR). IC31 gate 4 unused.
4. J2 = 40-pin IDE header (pin 20 kept) for a CF-to-IDE adapter: plugged straight on (TAODAN CF-IDE40, standing off the component side) or on a short ribbon (SinLoon). Adapter power: J3 (1 = +5 V, 2 and 3 = GND, 4 not used) or, with JP2 fitted, IDE pin 20 for adapters powered there. CSEL = GND (master).
5. Pull-ups: CF D0–7 (RN9; an empty adapter reads $FF = 'no card' to the ROM driver), IORDY, -PDIAG, -DASP: -DMACK held inactive. SRST (IC32 Q4) is a one-pin net by design (a probe point). -CFRESET = -RESET AND -SRST.
6. ACT LED: any P9 access (the buffer enable -CFOE = -IOR AND -IOW; ACTK sinks it). No PWR LED here: the card has PWR0 + R2 (sheet 1). No DASP LED on v2.0. C25 IC30, C26 IC31, C27 IC32, C28 IC33, C29 IC34; C30 10 uF at J3.
7. From hardware/cards/cf/kicad/v1.0/cf_netlist.py: U1->IC30 U2->IC31 U3->IC32 U4->IC33 U5->IC34 J1->J2 J2->J3 JP1->JP2 RN1->RN9 R1–R5->R10–R14 LED2->LED1 C1–C6->C25–C30. Dropped: X1 (the card's bus connector), the PWR LED (LED1, R7: the card has PWR0 + R2) and the DASP LED (LED3, R6).

Theory: docs/cards/cf.md, docs/cards/memory.md. Circuit source: mem_v2_netlist.py (edit, then re-run build.sh).
Drawn by tools/kicad/sch_wire.py: ELK places the parts and the section's own nets are wires (PWR_FLAGs: sheet 1).

Circuit source: mem_v2_netlist.py; this sheet is generated by gen_mem_v2.py
The CF card v1.0 circuit minus its bus connector, PWR LED and DASP LED
CF (True IDE, 8-bit) on I/O ports P8 (register latch) / P9 (data) – docs/cards/cf.md
YACC1

Sheet: /Sheet 7/		
File: memory-v2.0-sheet7.kicad_sch		
Title: memory-v2.0 sheet 7: CompactFlash interface		
Size: A2	Date: 2026-09-24	**Rev: v2.0**
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